

Quiz 1 Review

Question 1

This question concerns the hospital-resident problem (RHP). Assume there are n hospitals and n residents, and every hospital has exactly one position. In a natural variant, called RHPI, preference lists may be incomplete: a hospital does not have to rank every resident, and a resident does not have to rank every hospital. A hospital cannot be matched with a resident who is missing from its list, and a resident cannot be matched with a hospital that is missing from theirs.

Before stating what a good RHPI solution is, it helps to list the instabilities that can occur in a valid RHPI solution and that stop it from being good.

(a) Complete the definition below by selecting every phrase that belongs in it.

One possible instability is a pair (h, r') in which resident r' is unmatched and ...

1. hospital h appears on r' 's preference list;
2. hospital h prefers its current resident r to r' ;
3. hospital h prefers r' to the resident it is currently assigned;
4. hospital h is also unmatched.

(b) Describe how to build an instance with n hospitals and n residents, for any sufficiently large even n , in which every hospital's and every resident's preference list has length 2, and every valid solution matches exactly $n/2$ residents.

Question 2

Imagine that each scenario below arises when the Gale–Shapley algorithm is run on n employers and n applicants, with the employers proposing. For each one, give a good asymptotic lower bound $\Omega(\cdot)$ on the number of iterations of the while-loop. Leave out low-order terms and irrelevant constant factors. If the answer involves a logarithm of any base, just write $\log n$.

Scenario	Lower bound on iterations
All applicants except $\log n$ of them end up with their least-preferred employer.	$\Omega(?)$
Every employer has the same preference list.	$\Omega(?)$
A group of $n^{4/3}$ applicants is ranked below all other applicants by every employer, although the employers order the members of this group differently.	$\Omega(?)$
30% of the employers and 30% of the applicants agree in advance that each of them will rank the members of the other group ahead of everyone else.	$\Omega(?)$

Question 3: Reductions — True or False

A computer scientist, X, wants to solve problem A. They design an algorithm that reduces A to a problem B for which an algorithm already exists. For an instance I of A, let $r(I)$ denote the instance of B that X's algorithm produces. Select all the true statements about X's solution.

1. If B can itself be solved by a reduction to a problem C, then A can be solved by reducing it directly to C, without passing through B.
2. For every instance I of A, the solution to the B-instance $r(I)$ produced by the reduction is identical to the solution to I.
3. Suppose the reduction turns an instance of A of size n into an instance of B of size $f(n)$, and the algorithm for B runs in $\Theta(g)$ time. Then X's solution to A runs in $\Omega(g(f(n)))$ time.
4. Every instance I of A can be reduced to some instance J of B; that is, for every I there exists an instance J of B such that $J = r(I)$.